
Introduction to Natural Language Processing (NLP)

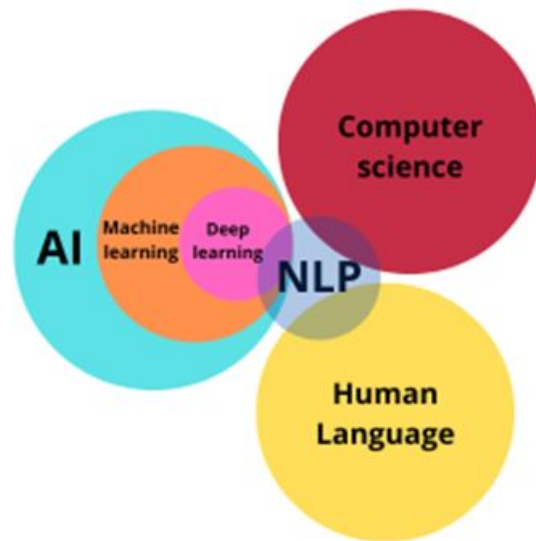
— Week 2 - NLP Class —

What is NLP?

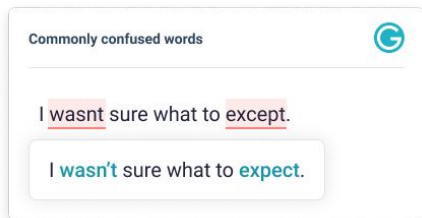


Natural Language is language that has developed in the usual way as a method of communicating between people → Human Language.

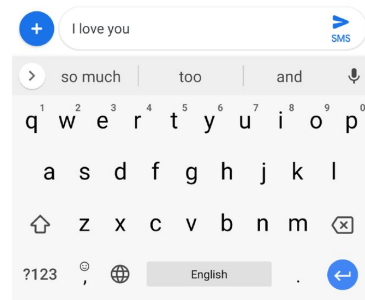
Natural Language Processing (NLP) is part of computer science and artificial intelligent (AI) which deals with human language.



Applications of NLP in the industry



Spell checking



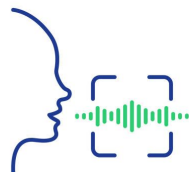
Word prediction



Keywords search



Sentiment analysis



Speech Recognition

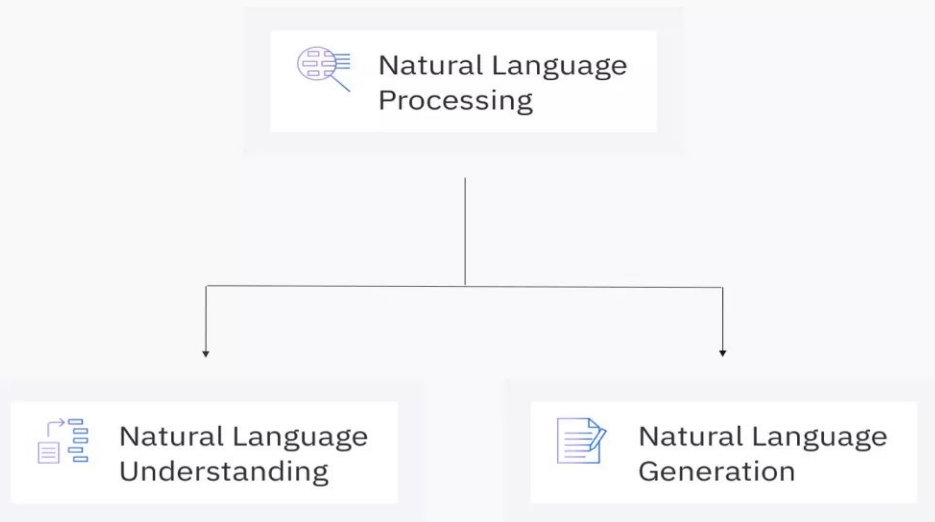


Chatbot

What else?

NLP Concepts

NLU uses **syntactic** and **semantic** analysis of text and speech to determine the meaning of a sentence. Syntax refers to the grammatical structure of a sentence, while semantics alludes to its intended meaning.



NLG focuses on text generation, or the construction of text in English or other languages, by a machine and based on a given dataset.

Read more about NLU and NLG: [NLU & NLG](#)

Challenge in NLP: Ambiguity

Ambiguity → the ability of **having more than one meaning** or **being understood in more than one way**.

Lexical ambiguity

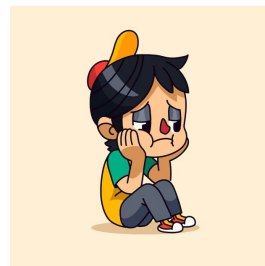


Syntactic Ambiguity



Referential Ambiguity

"The boy told his father about the bad news. **He** was very upset."



or



Read more about ambiguity: [Ambiguity](#)

How does a computer process text/human language?

Does a computer truly understand human languages such as English or French?

"She is a queen, and he is a king. They live happily in a beautiful castle."

"Elle est reine et il est roi. Ils vivent heureux dans un beau château."



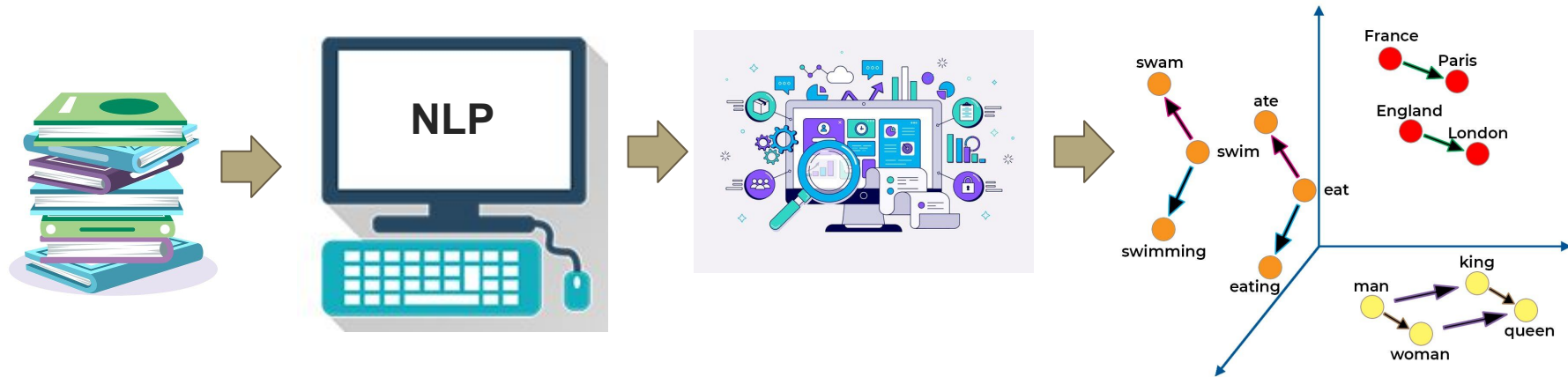
Long character string

Computer is very good at:

- Learning the patterns from data
- Doing a repetitive task

Computers don't understand language the way humans do. When a computer looks at text, it doesn't see words and sentences but sees a long string of characters. This is like giving someone a book in a language they don't understand. The symbols (or letters) have no inherent meaning to them.

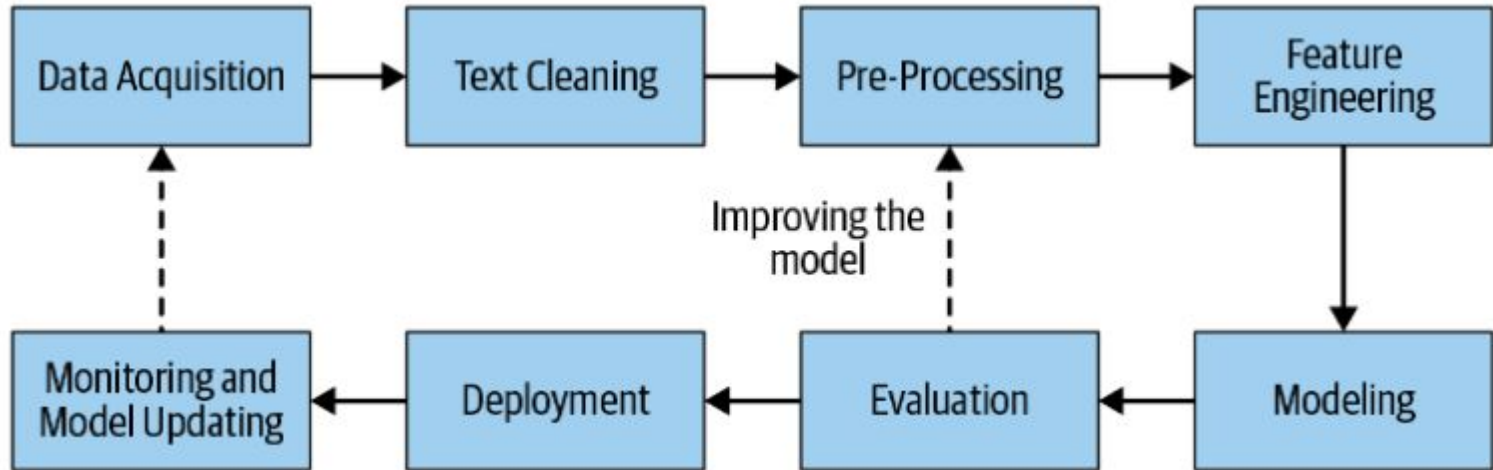
How does a computer process text/human language?



NLP is a set of tools and techniques that help computer find and learn patterns in text. For instance, every time it sees the word "dog", it often sees words like "bark" or "pet" nearby. Gradually, the computer starts to notice these patterns.

From the example above, the robot learns that "king" and "queen" are more closely related to each other than "king" and "swim". It's like teaching the computer that cats and dogs are both animals, but cats are more similar to lions than to fish.

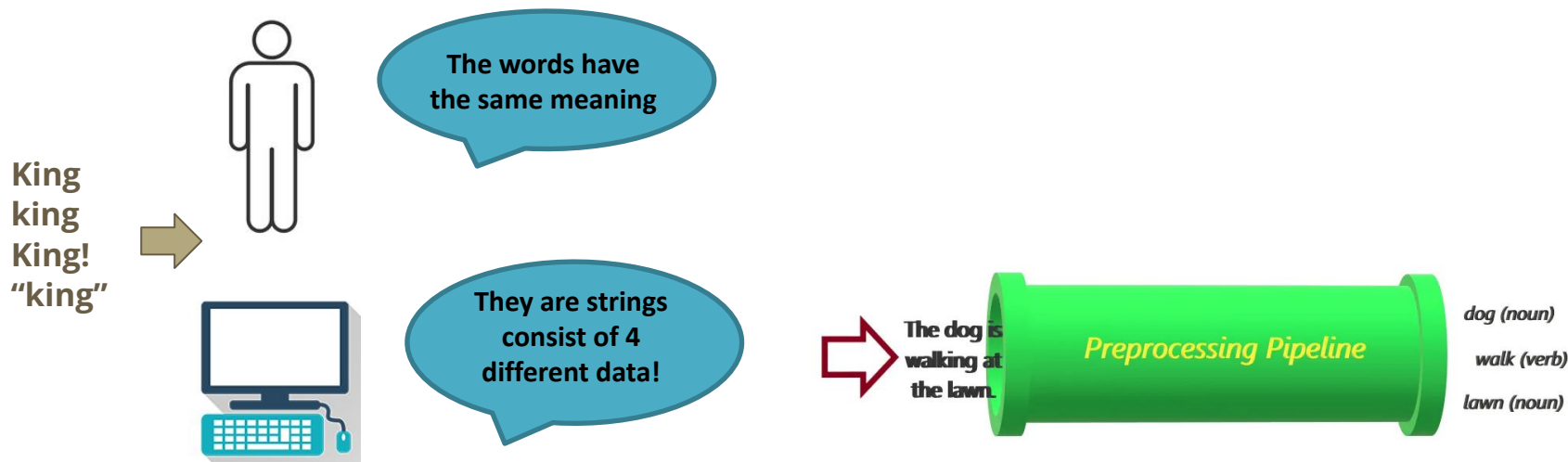
General end-to-end process for NLP



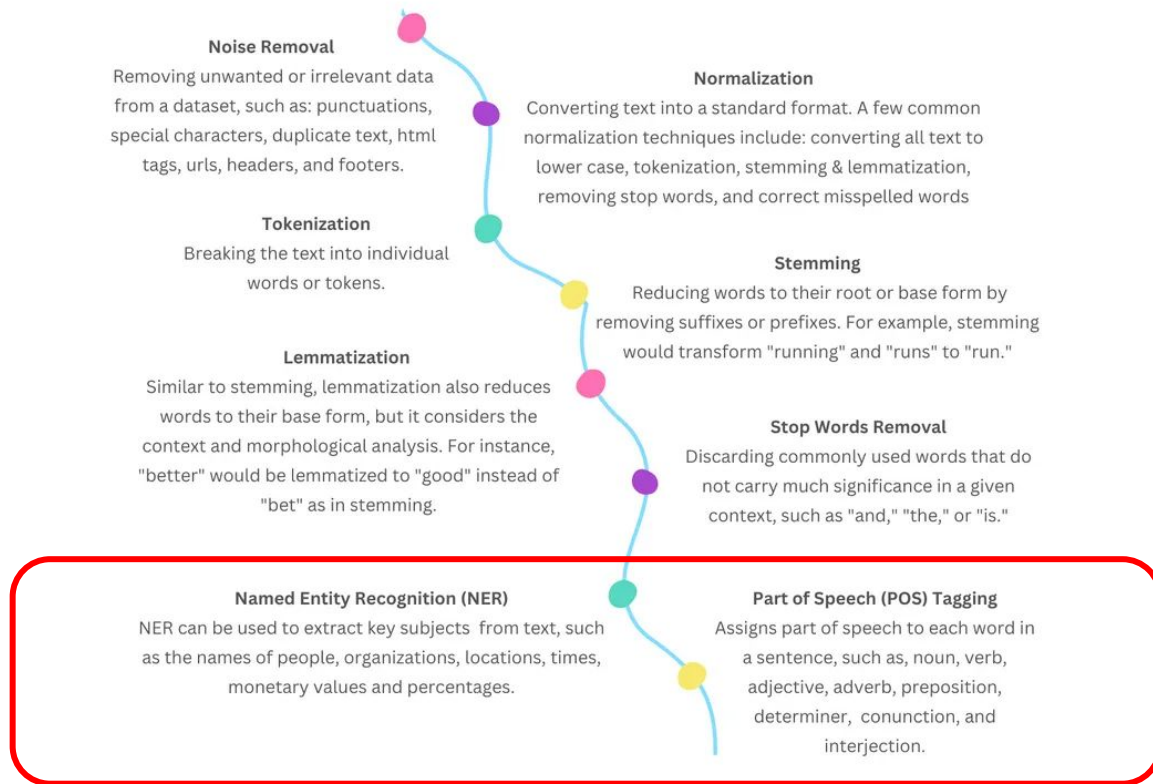
Text Cleaning and Pre-processing

Why?

Because computers don't understand language the way humans do.



Text Cleaning & Pre-processing



It's coding time!

Week 2 - Text preprocessing coding practice

